CS1111 — Problem Solving using Computers

Tier	A
Phase (order)	1
Do this first	Nothing
Estimated effort	6-8 weeks, ~6 h/week
Unlocks	CS1234, CS2705, CS2222

Roadmap

1. Install gcc, make and gdb on Linux (you already have Linux).
2. Work through the C basics in the order of the course's own weeks: variables, loops, arrays, strings, 2-D arrays, functions, recursion.
3. Solve each course-style problem (weighted marks, encrypt/decrypt a string, tic-tac-toe) yourself before looking at any solution.
4. Finish with pointers and structs, since every later course (OS, compilers, systems) assumes them.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Introduction to C Programming (60 videos). IIT Madras's own C course (BS Electronic Systems). Same style as your IITM course.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- CS50x 2025 Lectures (13 videos). CS50x 2025: watch lectures 1-5 (C, arrays, algorithms, memory). Best-produced intro anywhere.
- Problem solving through Programming In C (61 videos). NPTEL *Problem Solving through Programming in C* (Anupam Basu, IIT Kharagpur).

⚡ Short videos for intuition and revision

- C Programming (100+ videos). Neso Academy C: short topic videos, good for revising one concept.
- Programming in C (100+ videos). Jenny's Lectures C (Indian-English, exam style).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–14	How a computer works, number representation, compilation. Install gcc/gdb, write and compile hello-world by hand.
2	15–22	First programs, variables, operators, control flow, loops. Solve 10 small loop problems (patterns, primes, digit sums).
3	23–27	Functions, runtime stack, recursion, scope. Write factorial/fibonacci/power recursively and iteratively.
4	28–38	Pointers, endianness, arrays, pointer arithmetic, strings. Re-implement strlen/strcpy/strcmp yourself.
5	39–49	struct/typedef/union/enum, stack vs heap, malloc/free, 2-D arrays. Build a student-records program with a dynamic array.
6	50–60	Files, bit manipulation, macros, multi-file projects. Final: tic-tac-toe + caesar cipher split across .c/.h files with a Makefile.

▼ **All lectures in *Introduction to C Programming* (60 videos)**

1. 1.1 : How a Computer works
2. 1.2 : Generalized Memory and Computations
3. 1.3 : Split Memory Architecture
4. 1.4 : Loops, Conditions & Branches
5. 1.5 : Communication between Computer Memory and Outside World
6. 1.6 : What is Programming?
7. 1.7 : Introduction to C Language
8. 2.1 : Number Representation in a Computer
9. 2.2 : Negative Number and Hexadecimal Representation
10. 2.3 : Floating Point Representation
11. 2.4 : Character Representation and Encoding
12. 2.5 : Instructions Encoding
13. 2.6 : Program Compilation
14. 2.7 : Role of Operating System in a Computer
15. 3.1 : Writing First C Program
16. 3.2 : Structure of a C program
17. 3.3 : Variables in C
18. 3.4 : Operators in C
19. 4.1 : Structured Programming
20. 4.2 : Introduction to Control Flow in C
21. 4.3 : Control Flow in C - Examples
22. 4.4 : Loops in C
23. 5.1 : Why Functions?
24. 5.2 : Functions in C

25. 5.3 : What is a Runtime
26. 5.4 : Recursion
27. 5.5 : Scopes in C
28. 6.1 : Introduction to Pointers
29. 6.2 : Endianness
30. 6.3 : Alignment
31. 6.4 : Pointers & Variables
32. 6.5 : Compiler Optimizations
33. 7.1 : Introduction to Arrays - Part I
34. 7.2 : Introduction to Arrays - Part II
35. 7.3 : Pointer Arithmetic
36. 7.4 : Examples of Pointer Arithmetic
37. 7.5 : Strings in C
38. 7.6 : String Functions
39. 8.1 : Introduction to Struct in C
40. 8.2 : Examples of Struct - I
41. 8.3 : Examples of Struct - II
42. 8.4 : Typedef in C
43. 8.5 : Union in C
44. 8.6: Enum in C
45. 9.1 : Heap and Stack Frames
46. 9.2 : Dynamic Memory Allocation in C
47. 9.3 : Examples of Dynamic Memory Allocation - I
48. 9.4 : Examples of Dynamic Memory Allocation - II
49. 9.5 : Multidimensional Arrays in C
50. 10.1 : Introduction to File Handling in C
51. 10.2 : File Handling in Text Mode - I
52. 10.3 : File Handling in Text Mode - II
53. 10.4 : File Handling in Binary Mode
54. 10.5 : Bit Manipulation
55. 11.1 : Introduction to Macros in C
56. 11.2 : Examples of Macros - I
57. 11.3 : Examples of Macros - II
58. 11.4 : Include File in C
59. 11.5 : Multifile Projects in C
60. 11.6 : Compilation Process of C Programs

Books, notes and websites

- CS50x (Harvard), weeks 1-5 — cs50.harvard.edu/x
- Book: *The C Programming Language* (Kernighan & Ritchie) — also the CS2222 text

- Book: *C Programming: A Modern Approach* (K.N. King) — slower and gentler than K&R
- Beej's Guide to C — beej.us/guide/bgc (free)
- Practice: exercism.org C track, HackerRank C

Practice / proof of understanding

Write a tic-tac-toe, a caesar cipher and a matrix multiplication program in C with no copy-paste, compile with `-Wall -Wextra`, and run them under `valgrind`.

What your IITM course sheet says

From the PDF: CS1111

Official description: The course aims to provide exposure to problem-solving through programming. It aims to train the student to the basic concepts of the C programming language. This course involves a lab component which is designed to give the student hands-on experience with the concepts.

Official syllabus (trimmed):

> The course is structured around 8--10 Problems that will be solved throughout the semester. For each problem the necessary language constructs will be introduced. Week 1 and 2: Introduction and getting started Motivation of the course, fundamentals of computing, historical perspective, writing the first C program. Phases of developing and running a program in C. Statements, declarations, variables, basic data types (int, char), basic arithmetic and logical operators, if-else and loop construct. Week 3: Input from user (scanf), output to screen (printf), Arithmetic operators, precedence of operators. Sample Problem: Weighted sum of assignment marks for a single student. Week 4: Introduction to arrays, use of loops, need for different loop structures like while, do-while Sample Problem: Weighted sum of assignment marks for several students. Week 5: Introduction to character arrays, manipulating character arrays, motivate need for functions. Sample Problem: Encrypt and Decrypt a string. Week 6: 2-D arrays, use of loops, user input, input validation. Sample Problem: Tic-Tac-Toe program (2 users) Week 7: Functions. The prototype declaration, Function definition. Function call - Passing arguments to a function, by value, by reference. Scope of variable names. Week 8: Recursion, recursive formulas, factorial, Hemachandra (aka Fibonacci) numbers, recursive functions in C, sum, max of i

Official textbooks: C: How to program, H. M. Deitel, P. J. Deitel, 7th edition, Pearson Education, 2010.

Official references: R. G. Dromey, "How to Solve It By Computer", Pearson, 1982 A.R. Bradley, "Programming for Engineers", Springer, 2011 A.G. Ranade, "An Introduction to Programming through C++", McGraw Hill, 2014 Kernighan and Ritchie, "The C Programming Language", (2nd ed.) Prentice-Hall, 1988

From the PDF: CS1100

Official description: The students should be able to write basic C programs comfortably. They should develop the logic building ability to solve real word problems.

Official syllabus (trimmed):

> Module 1 : (Introduction to Computing) - 6 lectures Fundamentals of Computing, Historical perspective, Early computers. Computing machine. Problems, Pseudo-code and flowcharts. Memory, Variables, Values, Instructions, Programs. Module 2 : (Introduction to C) 10 lectures The language of C : Phases of developing a running computer program in C. Data concepts in C :Constants, Variables, Expressions, Operators, and operator precedence in C. Statements : Declarations, Input-Output Statements, Compound statements, Selection Statements. Conditions, Logical operators, Precedences. Repetitive statements, While construct, Do-while Construct, For construct. Data types, size and values. Char, Unsigned and Signed data types. Number systems and representations. Constants, Overflow. Arrays. Strings. Multidimensional arrays and matrices. Module 3 : (Modular Programming and Example Problems) : 10 lecture Functions :The prototype declaration, Function definition. Function call : Passing arguments to a function, by value, by reference. Scope of variable names. Recursive function calls, Tail recursion. Analysing recursion, Tree of recursion, linear recursion. Sorting problem : Selection Sort, Insertion Sort, Comparison between sorting algorithms. Sorting in multidimensional arrays. Sorting in strings. Search problem : Linear search and binary search. Comparison between search pro

Official textbooks: 1. Text Book 1: C: How to Program, P. Dietel and H. Dietel, 7th edition, Pearson Education, 2010

Official references: 1. Reference Book 1: Computer Programming in C, V. Rajaraman, 2nd edition, PHI Learning, 2008 2. Reference Book 1: How to solve it by Computer?, R. G. Dromey, 2nd edition, Pearson Education, 2008 3. Reference Book 1: The C Programming Language, B. W. Kernighan and D. M. Ritchie, 2nd Edition, Prentice Hall, 1998